

Universal collective Larmor-Silin mode emerging in magnetized correlated Dirac fermions

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Employing large-scale quantum Monte Carlo simulations, we find that in the magnetized interacting Dirac fermion model there emerges a universal collective Larmor-Silin spin wave mode in the transverse dynamical spin susceptibility. Such mode purely originates from the interaction among Dirac fermions and distinguishes itself from the usual particle-hole continuum with finite lifetime and clear dispersion, both at small and large momenta in a large portion of the Brillouin zone. Our unbiased numerical results offer the dynamic signature of this collective excitation in interacting Dirac fermion systems, and provide experimental guidance for inelastic neutron scattering, electron spin resonance, and other spectroscopic approaches in the investigation of such universal collective modes in quantum moiré materials, topological insulators, and quantum spin liquid materials under magnetic field, with quintessential interaction nature beyond the commonly assumed noninteracting Dirac fermion or spinon approximations.

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I. INTRODUCTION

Collective excitations offer the fingerprint of quantum many-body systems, e.g., the spin wave in the magnetically ordered systems [1–5], the roton mode in the superfluid [6–11], excitons and magnetorotons in the quantum moiré materials, integer and fractional quantum (anomalous) Hall systems [12–19], and many others. And it is oftentimes the case that the identification of new collective excitations provides the decisive understanding of the physical nature of the corresponding quantum many-body ground states.

The situation becomes subtle in highly entangled quantum matter, where the collective excitations are harder to identify in an unbiased manner. For example, in the study of quantum spin liquid (QSL) states, it is well known that the identification of the unique aspect of a QSL state from a specific experimental signature is difficult, as there usually exists multiple explanations for the same experimental data [20,20–22]. One crucial property of QSL is the emergence of fractionalized excitations, such as spinons (carrying a spin of 1/2 but charge neutral), which are elementary quasiparticles carrying topological characteristics and interacting with an emergent gauge field [23–31]. In Dirac QSLs, spinons can exhibit a conical dispersion [32], resembling Dirac cones in the electronic band structures of graphene, topological insulators, and many two-dimensional (2D) materials [33–36].

While detecting single spinon excitations is challenging, their collective excitations—two spinon excitations with a total spin quantum number $S = 1$ —can lead to a spin continuum spectrum that can be detected through inelastic neutron scattering techniques. Such characteristic collective continuum spectra have been reported in materials such as the kagome lattice antiferromagnets $\text{ZnCu}_3(\text{OH})_6\text{Cl}_2$ [37], $\text{Cu}_3\text{Zn}(\text{OH})_6\text{FBr}$ [38], and more recently $\text{YCu}_3(\text{OH})_6\text{Br}_2[\text{Br}_{1-x}(\text{OH})_x]$ [32,39–42].

However, in these studies of QSL, one often assumes the fractionalized spinons are nearly free particles and computes their collective modes (continuum spectrum) under such assumption, i.e., the convolution of two independent spectra of a single spinon [43–45]. But in reality, it is obvious that the spinons experience strong interactions, mediated by the fluctuations of gauge fields [27–29,46–49], and the free spinon assumption is oversimplified and often leads to contradictions or controversies, when trying to interpret the experimental data and make predictions [50,51].

Therefore, one needs to either solve the interacting problem completely, usually via unbiased numerical approaches such as quantum Monte Carlo (QMC) [26–31,48], or find new signatures which are robust beyond the nearly free approximations, even when the interaction effect is included. But neither of these *two strategies* is easy and the progresses are usually made in a case by case manner. Recently, an interesting proposal of the latter strategy came to our attention. In Refs. [52–54], the authors propose a new collective “spinon spin wave” mode to emerge in the transverse dynamical spin susceptibility—different from the usual spin

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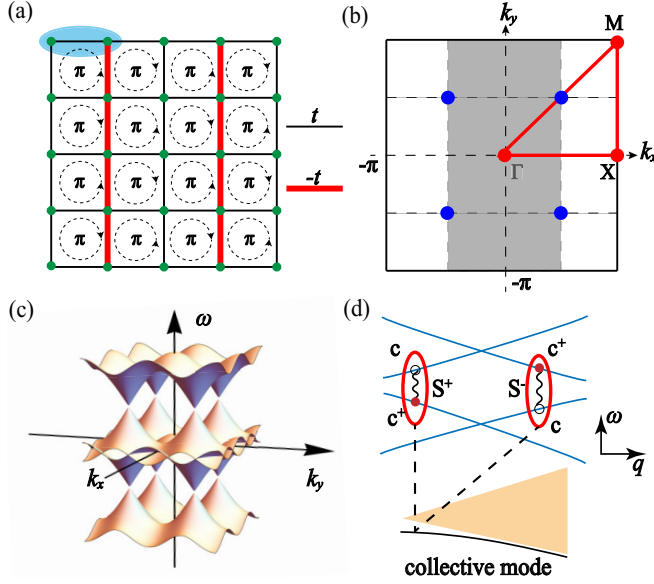


FIG. 1. (a) Schematic plot of the Hamiltonian Eq. (1), the π -flux model on the square lattice. The black lines stand for the hopping sign of the fermion with $t_{ij} = +1$, while the red lines are $t_{ij} = -1$. Such arrangement ensures that the flux threading each plaquette is π , giving Dirac cones at zero magnetic field. The blue ellipse enclosing two lattice sites is the unit cell. (b) BZ of the original square lattice (white) and the folded BZ (gray) with blue ellipse as unit cell. The path denoted by the red line connects high symmetry momenta $\Gamma(0, 0)$, $X(\pi, 0)$, and $M(\pi, \pi)$ in BZ for the original square lattice. (c) The split Fermi surface of the π -flux model under external magnetic field in z direction. (d) Schematic plot that depicts the origin of the collective mode. The coherent excitation of spin flip S^+ and S^- in the model gives rise to the collective mode.

continuum spectra—in the 2D Dirac QSL subjected to an applied Zeeman field, and tested their proposal rigorously in a one-dimensional Heisenberg chain via density matrix renormalization group simulations and in 2D magnetized graphene in perturbative analysis.

The suggested collective modes date back to early investigations of the spin response of paramagnetic metals subjected to the external magnetic field, denoted as the transverse Larmor-Silin spin wave [55,56], in the form of collective spin oscillations in itinerant interacting fermion systems [57–60]. Phenomenologically, such Larmor-Silin spin wave is the transverse collective spin mode below the particle-hole continuum of, say, a single-band conductor with parabolic electron dispersion. This downward dispersing collective mode is the precession of the total magnetic moment originating from the Zeeman frequency at zero wave vector [as schematically shown in Fig. 1(d)] and of purely interacting nature. The spinon spin wave in Refs. [52–54], can therefore be viewed as the modern version of the Larmor-Silin spin wave, in the context of magnetized graphene and 2D Dirac QSL.

Until now, the unbiased 2D lattice model verification of the spinon spin wave, i.e., to carry out both the strategy of solving the interacting problem completely via unbiased numerical calculation and the strategy of identifying a new signature of the purely interaction-generated collective mode beyond

the nearly free approximation, has not been achieved. This is due, of course, to both the lack of proper lattice model design and the numerical difficulties in solving the interacting problem accurately. In this Letter, we finally achieve both goals successfully, by employing large-scale QMC simulations, complemented with random phase approximation (RPA) analysis, to find that in a concrete 2D magnetized interacting Dirac fermion lattice model there emerges the universal collective Larmor-Silin spin wave mode in the transverse dynamical spin susceptibility. Beyond the perturbative proposals [53,54], we find in the 2D lattice model that the Larmor-Silin spin wave not only splits off from the usual “two-spinon continuum” at small momenta, but also universally appears inside and at the bottom of the continuum at large momenta, with richer renormalization effects and magnetic field dependence. Our unbiased numerical results offer the dynamic signature of this collective mode in interacting Dirac fermion systems, and might provide experimental relevance for its detection via spectroscopic measurements (for example inelastic neutron scattering) in quantum moiré materials, topological insulators, and Dirac spin liquid materials, under magnetic field.

II. MODEL AND METHOD

Our model has the following Hamiltonian on a 2D square lattice:

$$\begin{aligned}
 H &= H_0 + H_U + H_B, \\
 H_0 &= -t \sum_{\langle i,j \rangle, \sigma} (t_{ij} c_{i,\sigma}^\dagger c_{j,\sigma} + \text{H.c.}), \\
 H_U &= \frac{U}{2} \sum_i (n_{i\uparrow} + n_{i\downarrow} - 1)^2, \\
 H_B &= -B \sum_i \frac{1}{2} (c_{i,\uparrow}^\dagger c_{i,\uparrow} - c_{i,\downarrow}^\dagger c_{i,\downarrow})
 \end{aligned} \tag{1}$$

and as shown in Fig. 1(a), the hopping sign t_{ij} on black (red) bonds is $+1$ (-1), ensuring the flux threading each plaquette is equal to π . $\sigma = \uparrow, \downarrow$ denotes spin flavors. The blue ellipse denotes the unit cell, which contains two sites. The Brillouin zone (BZ), shown in Fig. 1(b), is folded with $k_x \in [-\pi/2, \pi/2]$ and $k_y \in [-\pi, \pi)$. The free dispersion has two independent Dirac points denoted by the blue solid points. Near the Dirac points, the dispersion has good linear form. The interaction in the model is the on-site Hubbard repulsion with strength U . The $S_i^z = \frac{1}{2}(c_{i,\uparrow}^\dagger c_{i,\uparrow} - c_{i,\downarrow}^\dagger c_{i,\downarrow})$ of the fermion is coupled to an external magnetic field in z direction with strength B . The effect of magnetic field B can be interpreted as that the spin up and spin down fermion have chemical potential with opposite sign.

Without external magnetic field, the π -flux Hubbard model with spin 1/2 fermion has a Dirac semimetal to antiferromagnetic Néel state quantum phase transition at finite U_c , with the latter state spontaneously breaking $SU(2)$ spin rotational symmetry [61–65]. In the presence of Zeeman field B , there will also be a paramagnetic to in-plane antiferromagnetic state transition at $U_c(B)$ [61]. However, since we are interested in the Larmor-Silin mode as a collective transverse spin excitation without spontaneous magnetic order, our main results

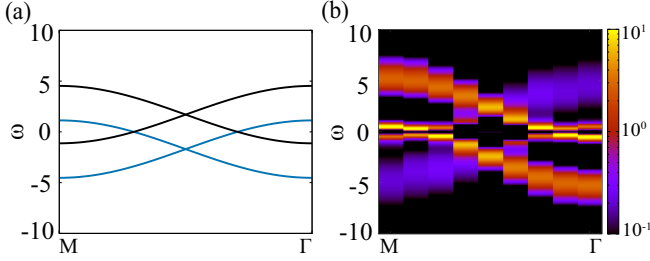


FIG. 2. (a) Free fermion dispersion under magnetic field $B = 3.4$. Because of polarization, the spin degeneracy is lifted and four bands are visible. (b) Single particle spectra $A(k, \omega)$ from QMC+SAC with parameter $U = 2, B = 3.4$ on a 16×16 lattice. The gap at $(\frac{\pi}{2}, \frac{\pi}{2})$ is around 2.34, corresponding to an effective magnetic field $B_{\text{eff}} = 4.68$ in Eq. (2).

are in the regime of $U < U_c(B)$ such that the spontaneous in-plane antiferromagnetic state has not been established but the magnetized Dirac fermions are nevertheless interacting. The magnetic spectra in the situation of $U > U_c(B)$ are presented in the Supplemental Material (SM) [66] (see also Refs. [67,68] therein). With plaquette interactions, extended beyond on-site, it is also found via large-scale QMC simulation that the model can host a $U(1)$ Dirac QSL phase, via the nontrivial deconfined quantum critical point [69–72]. It is fair to assert that the π -flux Dirac fermion model can be used to describe the spinon dispersion of a $U(1)$ Dirac QSL [73]. Therefore, the spin spectra of our model can help to understand the spectrum of the $U(1)$ Dirac QSL under magnetic field. Admittedly, the present model does not have the dynamical $U(1)$ gauge field and we shall leave it for future work.

III. RPA ANALYSIS

First we investigate the model Eq. (1) with RPA following the work [53]. The detailed calculation scheme is given in Sec. I of SM [66]. We define magnetization as $M = \frac{1}{2}(\frac{N_\uparrow - N_\downarrow}{N_\uparrow + N_\downarrow})$, where $N_\uparrow$ ($N_\downarrow$) is the particle number of the up (down) fermion. The system is at half filling, $N_{\text{tot}} = N_\uparrow + N_\downarrow = L^2/2$, where L is the linear system size. The contribution to the Green's function from Hubbard interaction U is approximately treated as Hartree energy shift. The fermion dispersions are shifted by $-UM$ for up spin and UM for down spin. In practice, we find it more appropriate to use renormalized interaction strength U_{eff} in RPA calculation. As shown in Fig. 2, in QMC we can obtain the effective magnetic field B_{eff} which is two times the gap at the Dirac point $(\pi/2, \pi/2)$ from single particle spectra $A(\mathbf{k}, \omega)$. B_{eff} is then related to U_{eff} by

$$B_{\text{eff}} = B + U_{\text{eff}}M. \quad (2)$$

The renormalized interaction strength U_{eff} is then calculated as $U_{\text{eff}} = \frac{B_{\text{eff}} - B}{M}$. Consider the effective Hamiltonian, $\epsilon_{a,\sigma}(k) = (-1)^a 2t \sqrt{\cos^2 \mathbf{k}_x + \cos^2 \mathbf{k}_y} - \alpha(\sigma) B_{\text{eff}}/2$, where $a \in [1, 2]$ denotes the band index and $\alpha = +1(-1)$ for $\uparrow$ ($\downarrow$) spin. We then calculate the bare transverse spin susceptibility $\chi_{\pm}^0$ with standard fermion loop diagram. The RPA spin susceptibility $\chi_{\pm}$ is obtained by resummation of particle-hole

ladder diagrams with $\chi_{\pm}^0$ as bare susceptibility [74]:

$$\chi_{\pm}(\mathbf{q}, i\omega_n) = \frac{\chi_{\pm}^0(\mathbf{q}, i\omega_n)}{1 + U_{\text{eff}}\chi_{\pm}^0(\mathbf{q}, i\omega_n)}. \quad (3)$$

After analytic continuation $i\omega_n \rightarrow \omega + i0^+$, the spin spectra $S_{\pm}(\mathbf{q}, \omega) = -\text{Im}\chi_{\pm}(\mathbf{q}, \omega)$, which will be directly compared with QMC results. The existence of the universal collective mode is already manifested at RPA spin susceptibility [53,74] and the condition for its emergence is discussed in detail in SM [66].

IV. QMC SIMULATION AND RESULTS

We further investigate the model with finite-temperature determinant quantum Monte Carlo (DQMC) simulations [75,76]. The model in Eq. (1) is particle-hole symmetric and thus sign-problem free [77,78]. We use the Hubbard-Stratonovich decomposition to decouple the interaction into the density channel and perform the simulations on a 16×16 square lattice (at the inverse temperature $\beta t = L$ with $t = 1$ as the energy unit) at $U = 2$ with varying external magnetic field B . The external magnetic field B enters the DQMC as the chemical potential term for canonical ensemble simulations, with opposite sign for opposite spin flavor, and still renders the simulation sign-problem free. As discussed above, at large $U > U_c(B)$, the system will break the $U(1)$ spin rotational symmetry and enter an antiferromagnetic XY (AFXY) ordered insulator phase [61], but we purposely stay in the paramagnetic phase to investigate the transverse Larmor-Silin spin wave mode. The details of DQMC implementation and the magnetic spectra at $U > U_c(B)$ are given in Secs. II and III of SM [66], respectively.

First we calculate the imaginary time Green's function $G_{\sigma}(\mathbf{k}, \tau)$ and use the stochastic analytic continuation (SAC) [1,26,27,29,49,79–81] to obtain the real frequency single particle spectral function $A(\mathbf{k}, \omega)$. The result of $A(\mathbf{k}, \omega)$ is shown in Fig. 2(b) comparing with dispersion of the free fermion under the same external magnetic field in Fig. 2(a). As shown in the figure, the degeneracy of energy bands is lifted by the external magnetic field B , giving rise to the pocketlike Fermi surface shown in Fig. 1(c). In our lattice simulation with $L = 16, \beta = 16, U = 2, B = 3.4$, the QMC single particle spectrum shows that the magnetized system is still in the metallic state, and the amount of energy shift is larger compared to the free system with $U = 0, B = 3.4$, which can be attributed to the renormalization effect of the Hubbard interaction. This further confirms the usage of effective Hubbard U_{eff} as Hartree energy shift in RPA calculation.

Next we focus on the spin spectra of the magnetized Dirac fermions. We compute the dynamical transverse spin correlation function

$$S_{\pm}(\mathbf{q}, \tau) = \frac{1}{L^4} \sum_{i,j} \langle S_i^{\pm}(\tau) S_j^{\mp}(0) \rangle e^{-i(\mathbf{r}_i - \mathbf{r}_j) \cdot \mathbf{q}}, \quad (4)$$

where $S_i^+ = c_{i,\uparrow}^{\dagger} c_{i,\downarrow}$ and $S_i^- = c_{i,\downarrow}^{\dagger} c_{i,\uparrow}$. The same as $A(\mathbf{k}, \omega)$, we can obtain the spin spectra $S_{\pm}(\mathbf{q}, \omega)$ from $S_{\pm}(\mathbf{q}, \tau)$ utilizing SAC, whose details are given in Sec. II of SM [66].

The results are depicted in Fig. 3. The upper panels are RPA spin spectra $S_{\pm}(\mathbf{q}, \omega)$, while the lower panels are QMC

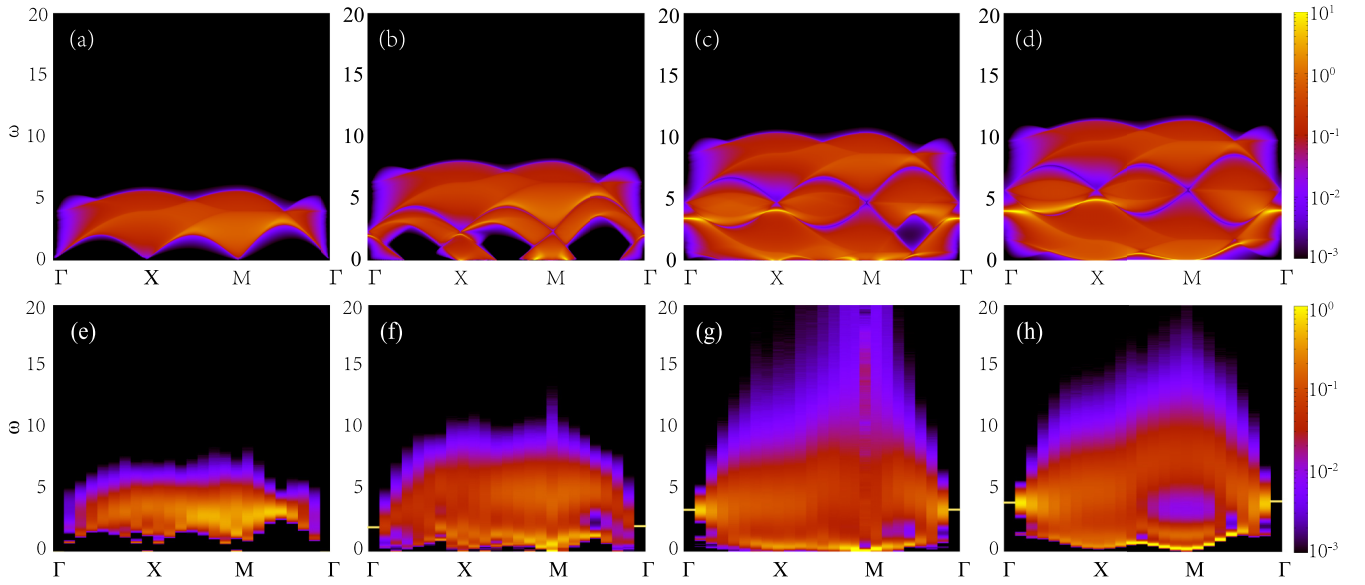


FIG. 3. Transverse magnetic spectra $S_{\pm}(\mathbf{q}, \omega)$ under z direction magnetic field at $U = 2$. (a)–(d) Spectra from RPA spin susceptibilities $\chi_{\pm}(\mathbf{q}, \omega)$ with $B = 0, 2, 3.4, 4$. (e)–(h) Spectra from QMC+SAC results with $B = 0, 2, 3.4, 4$. As magnetic field B increases, the universal collective transverse mode emerges, not only close to Γ but also at higher momenta and inside the spin continuum. The RPA and QMC spin spectra are consistent with each other.

results, with the parameters $U = 2, L = \beta = 16$. The RPA calculations are at zero temperature with 50×100 momentum grid; considering the unit cell, the BZ is folded with $k_x \in [-\pi/2, \pi/2]$ and $k_y \in [-\pi, \pi]$. The effective U_{eff} in the RPA panels are determined according to Eq. (2) with the help of single particle spectra $A(\mathbf{k}, \omega)$. For $B = 2, 3.4, 4$, effective $U_{\text{eff}} = 2.86, 1.69, 1.67$ respectively. Without magnetic field at $U = 2$, the magnetization $\langle M \rangle = 0$ in QMC, thus use $U_{\text{eff}} = U = 2$ in RPA.

In Fig. 3, we increase the magnetic field from left to right. At $B = 0$, we have the well-known spin continuum spectra also with Dirac cones at momenta Γ, X , and M [49]. Note that without magnetic field, the RPA spin spectrum has its weight redistributed compared with free spin spectra, due to the presence of interaction [49]. With increasing B , the lower part of the spectra appears from the vacuum, making the emergence of the Larmor-Silin spin wave collective mode possible. For both RPA and QMC results at $B = 2, 3.4, 4$, we can observe the emergence of quasiparticlelike excitation alongside the continuum. Specifically in Fig. 4(a), each momentum $\mathbf{q}$ of $S_{\pm}(\mathbf{q}, \omega)$ at $U = 2, B = 4$ [corresponding to Fig. 3(h)] is plotted such that we can clearly see the difference between continuum and collective mode excitations in QMC+SAC spectra. With larger B , there are more momenta possessing the universal collective mode, with sharp dispersion and finite lifetime.

At small momenta $\mathbf{q} \sim \Gamma$, the collective mode appears at around $\omega \sim B$. And it can be shown that at exactly the Γ point, the RPA treatment with effective U_{eff} is exact, meaning that the collective mode should be exactly at $\omega = B$ [53], independent of Hubbard interaction U . This is verified by QMC results in lower panels of Fig. 3 with different magnetic fields and fixed $U = 2$. On the other hand, we also fix magnetic field strength and vary U in QMC. The imaginary time spin correlation

functions at $\mathbf{q} = \Gamma$ for different U with the same magnetic field $B = 4$ are shown in Fig. 4(b). They all have the same slope in logscale plot, indicating identical spin excitation gaps whose value equals to B . The mechanism behind it is the Larmor/Kohn theorem [82]; i.e., when $SU(2)$ rotational symmetry of a spin system is only broken by external Zeeman magnetic field, at small momentum there are collective transverse spin excitations at B . This is the Larmor-Silin spin wave [55,56] discussed in Refs. [52–54]. We note that even though the position of the collective mode at the Γ point is independent of U , this property as well as the emergence of the collective mode itself are purely interaction effects.

Moreover, we find that at large momentum $\mathbf{q}$, the collective mode resides at the lower boundary of the continuum, which is not discussed in Ref. [54]. Compared with the low-energy effective field-theory calculation, our results of the π -flux model are based on more realistic band structures, and our finding of low-energy collective modes at large momenta in

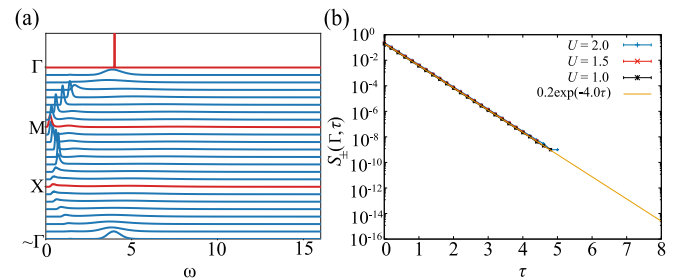


FIG. 4. (a) Spectra $S_{\pm}(\mathbf{q}, \omega)$ from QMC+SAC at $U = 2, B = 4$. The red lines highlight the high symmetry momentum points in BZ. (b) Logscale plot of imaginary time spin correlation function $S_{\pm}(\Gamma, \tau)$ from QMC simulation with $B = 4$ and varying U .

the BZ suggests that experiments can look for signatures of such collective modes in a large portion of BZ in realistic materials such as magnetized graphene and Dirac spin liquid candidates; we note that in inelastic neutron scattering experiments, very often the Γ point is blocked due to instrumental difficulties [32]. Furthermore, we notice that in RPA near $X(\pi, 0)$, there is a second collective mode roughly around B , while in QMC, a broader peak resides. The difference may be due to higher order contributions from the interaction term for larger momenta.

V. CONCLUSION

We investigate the spin dynamics of magnetized correlated Dirac fermions realized in a π -flux Hubbard model under Zeeman field, with RPA analysis and unbiased DQMC calculations. We find that when the Dirac fermions are interacting, there emerges a universal collective transverse spin mode—the Larmor-Silin mode—apart from the usual particle-hole continuum spectra. In particular, such collective mode not only splits off from the two-spinon continuum at small momenta as predicted in Refs. [52–54], but also appears inside and at the bottom of the continuum at large momenta, suggesting richer renormalization effects and magnetic field dependence in the lattice model. Our unbiased results can be used to guide inelastic neutron scattering, electron spin resonance, and other spectroscopic experiments of quantum moiré

materials, topological insulators, and spin liquid materials, under magnetic field.

As pointed out in Ref. [54], when the interaction is extended, other collective modes (such as the spin-current collective mode) would appear. It will be interesting to include the dynamical gauge field coupling with the Dirac fermion [28] to find these collective modes of magnetized Dirac spin liquid. We leave it for future work.

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